

**AI-BASED DETECTION OF FAKE JOB ADVERTISEMENTS
USING NATURAL LANGUAGE PROCESSING
WORDS COUNTS (10000)**

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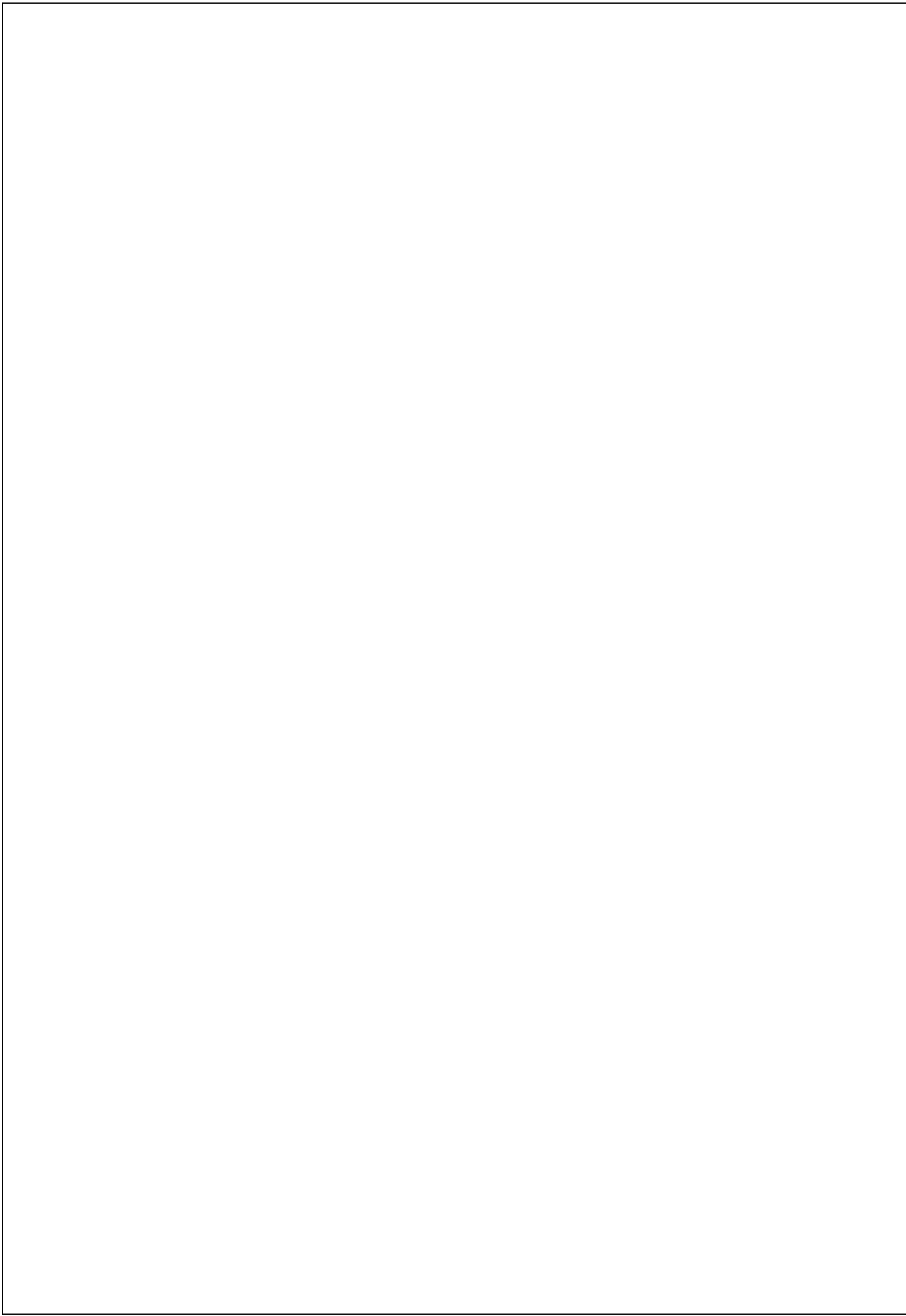
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In

Computing / Data Science / Web Development



Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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S. Younus

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Abstract

Job seekers can be defrauded for monetary losses and can have their identities stolen online, and job postings can be phony. An Artificial Intelligence (AI) framework which uses Natural Language Processing (NLP), machine learning and Explainable Artificial Intelligence (XAI) for fake job ads detection was developed. The literature surveyed recent research regarding the detection of recruitment fraud, representation of NLP features, supervised machine learning algorithms and explainability techniques. The proposed framework was divided in text preprocessing, feature extraction (TF-IDF), SMOTE for class imbalance and 4 machine learning algorithms which were namely Multinomial Naïve Bayes, Linear Support Vector Machine, Random Forest and Gradient Boosting. The results obtained from the experiments showed that the identified best model for this study is the Linear SVM model with good predictive performance and the model can be easily explained by SHAP and LIME. The results indicate that the framework is a feasible, interpretable and accurate solution for online recruitment fraud detection.

Table of Contents

Declaration.....	3
Acknowledgement	4
Abstract	5
List of Figures	8
List of Tables	9
Chapter 1: Introduction	10
1.1 Introduction	10
1.2 Problem Statement	10
1.3 Research Aim	10
1.5 Research Questions	11
1.6 Motivation	11
1.7 Research Contributions	11
1.8 Methodology Overview	12
1.9 Legal, Ethical and Professional Issues	13
1.10 Report Structure	13
Chapter 2: Literature Review and Technology Review.....	14
2.1 Introduction	14
2.2 Online Recruitment Fraud.....	14
2.3 Fake Job Advertisement Detection	15
2.4 NLP Feature Representation Techniques.....	16
2.5 Machine Learning Algorithms used for Detecting Fake Job Postings.....	17
2.6 Explainable AI.....	18
2.7 Research Gap.....	19
Chapter 3: Methodology and Implementation	21
3.1 Research Design.....	21
3.2 Development Environment	21

3.3 Dataset Description	22
3.4 Data Preprocessing and Quality Assessment	23
3.5 Exploratory Data Analysis	26
3.6 Feature Extraction	28
3.7 Handling Class Imbalance.....	29
3.8 Model Development and Evaluation Metrics.....	30
3.9 Explainable AI.....	32
3.10 Project Management.....	33
Chapter 4: Results and Evaluation.....	36
4.1 Experimental Setup	36
4.2 Analysis of Performance of the Models	36
4.3 Comparison with Previous Research.....	40
4.4 Critical Discussion	41
Chapter 5: Conclusion.....	42
5.1 Overall Conclusion.....	42
5.2 Achievement of Objectives	42
5.3 Research Contributions	44
5.4 Limitations	45
5.5 Future Work	45
5.6 Reflection	46
Reference List	48
Appendices.....	54
Appendix A: Proposal	54
Appendix B: GitHub Link.....	58
Appendix C: Video Link	59

List of Figures

Figure 1.1: Methodological Overview	12
Figure 3.1: Dataset Details.....	22
Figure 3.2: Missing Values and Target Class Distribution.....	24
Figure 3.3: Function for Cleaning Text Data.....	25
Figure 3.4: Applying Text Cleaning Operations.....	25
Figure 3.5: Applying Feature Engineering Operations.....	26
Figure 3.6: Target Class Distribution Visual	26
Figure 3.7: Description Length by Fraud Status	27
Figure 3.8: Correlation between Variables	28
Figure 3.9: Feature Extraction using TF-IDF	28
Figure 3.10: Using SMOTE to Handle Class Imbalance	30
Figure 3.11: Model Development and Evaluation.....	31
Figure 3.12: Performing Shap Analysis.....	32
Figure 3.13: Performing LIME Analysis	33
Figure 4.1: Performance of the Models	37
Figure 4.2: Model Comparison in terms of Precision-Recall AUC.....	37
Figure 4.3: Confusion Matrix	38
Figure 4.4: Feature Importance using Shap	39
Figure 4.5: Top Features Contributing to Fraud using LIME.....	39

List of Tables

Table 3.1: Gantt Chart.....	34
Table 3.2: Project Milestones and Risk Assessment	34
Table 5.1: Achievement of Research Objectives	43

Chapter 1: Introduction

1.1 Introduction

The rapid expansion of online recruitment platforms has transformed hiring by improving accessibility and efficiency for employers and job seekers. The same digital transformation has led to the propagation of fake job offers that have caused identity theft, economic losses and attacks by phishing that have caused a serious problem in cyber security. This requires automated and scalable detection solutions which are able to accurately detect deceptive recruitment content. Artificial Intelligence (AI) and Natural Language Processing (NLP) can be used to intelligently analyse text patterns, while Explainable Artificial Intelligence (XAI) can be used to make predictions interpretable to boost trust and accountability and the adoption of automated fraud detection systems [1].

1.2 Problem Statement

Job scams are fake job offers created to deceive the job seeker into providing personal information, paying up-front or conducting illegal financial transactions. As these scams have become more advanced, such as phishing mail, fake recruitment agencies, hallucinated employer websites and messaging apps like WhatsApp, it has become more difficult to spot them as they have become harder to detect manually. EHS provides recent reports from the U.S. According to the Federal Trade Commission, the total monetary loss reported from job scams in the United States during the first six months of 2024 was more than US\$220 million, an increase in the scale of job scams [2]. Traditional detection methods include key-word based detection and the classic machine learning algorithms that are not capable of understanding the context of deception and adapting to detect new fraud techniques. This puts organisations at risk of financial losses, reputational damage, regulatory risks and loss of user trust. Although the recent research has focused on enhancing the classification accuracy via deep learning and transformer-based approaches, research on data set quality assessment, class imbalance, and explainable decision making are lacking, which are crucial for the practical implementation and trustworthy recruitment fraud detection system [4].

1.3 Research Aim

The primary aim of this work is to create and evaluate an “*explainable Artificial Intelligence (XAI)*” based fraudulent job posting detection model in online recruitment platforms (marketplaces) based on “*Natural Language Processing (NLP) and machine learning techniques*”.

1.4 Research Objectives

- To preprocess and transform job advertisement text using “*Natural Language Processing*” techniques for machine learning analysis.
- To construct and contrast “*multiple machine learning*” models to detect fraud job ads.
- To measure the performance of a model using “*accuracy, precision, recall, F1 score and ROC-AUC*”.
- To use the methods of Explainable AI to discover important indicators that impact fraud decision making.

1.5 Research Questions

RQ1

- How effectively can Natural Language Processing techniques distinguish fraudulent job advertisements from legitimate postings?

RQ2

- Which machine learning algorithm has the best performance in the fake job detection?

RQ3

- What linguistic and contextual characteristics are most associated with fraudulent job advertisements?

RQ4

- How can “*Explainable AI*” improve the transparency and interpretability of fraud detection models?

1.6 Motivation

Fraudulent online recruitment has been propagated and a need for an intelligent detection system in both academic and practice aspects is felt. The work is both academically valuable and contributes to the emerging inter-disciplinary areas of Natural Language Processing, machine learning, Explainable Artificial Intelligence and cybersecurity, particularly regarding transparency in fraud detection within text data. Efficient detection systems can help recruitment platforms, companies and regulators minimize monetary losses, protect their reputation and gain consumer confidence from an industry standpoint. Socially, prevention of ID theft and financial exploitation of consumers leads to safer online work. In addition, improving the security of recruitment will enhance the overall cyber security resilience by reducing the constantly changing social engineering attacks.

1.7 Research Contributions

This dissertation makes a contribution to the study of recruitment fraud detection with a thorough analysis of a publicly available fake job advertisement dataset, data quality

assessment and class imbalance evaluation. It constructs and contrasts a number of machine learning models and new NLP feature representation techniques as well as identifies the best model to classify fraudulent job advertisements. Besides, the methods of Explainable Artificial Intelligence (XAI) are used to increase the transparency of the model and interpret the most relevant features and the meaning of the predictive model results. Finally, the study offers valuable recommendations to design dependable, understandable and scalable fraud detection systems to secure and trusted online recruitment.

1.8 Methodology Overview

Research methodology is a systematic approach that will be used for developing and evaluating proposed framework for fraud detection. First, a set of publicly available data of job ads is acquired and EDAs are applied to get insight into characteristics of the data, distribution of features and class imbalance. The data is then subjected to a process of preprocessing which involves both text cleaning and missing value handling and feature transformation. Later, the textual representations of the data are enriched with features of interest for the classification process (feature engineering).

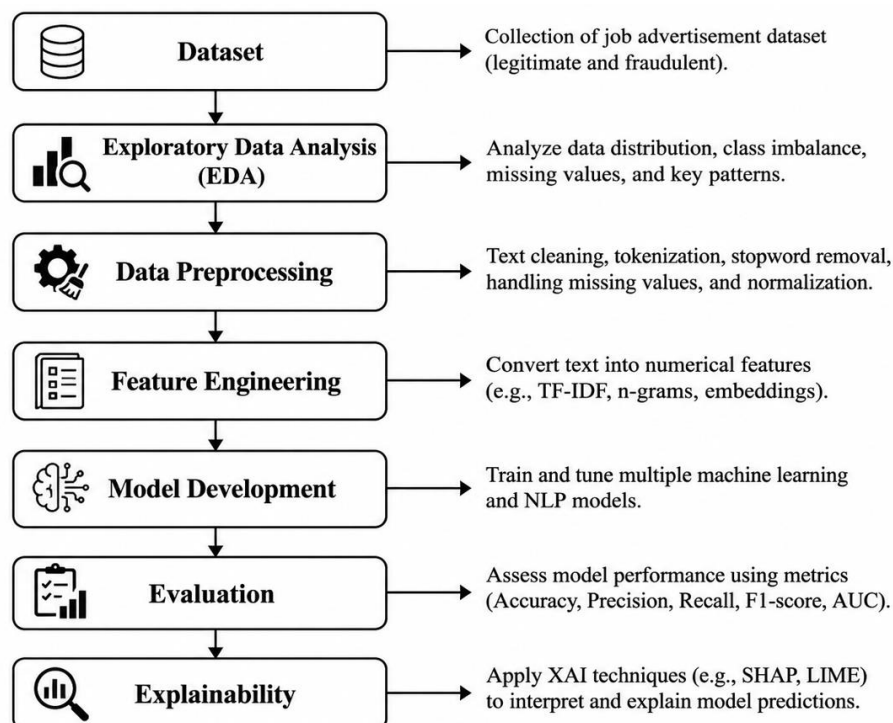


Figure 1.1: Methodological Overview

Various machine learning models are built and tested with the suitable performance measures. Finally, Explainable Artificial Intelligence (XAI) techniques are used to explain the model decisions, provide insight on the features affecting the model decisions, and increase the model transparency and trustworthiness.

1.9 Legal, Ethical and Professional Issues

Proposed research should follow legal, ethical and professional guidelines of data collection, model development, and model testing. Due to the use of a publicly released data set and the General Data Protection Regulation (GDPR), compliance with the licence conditions and to GDPR is crucial in assuring that any potentially identifiable information is processed lawfully. Responsible AI principles for non-discriminatory outcomes and trustworthy decision making include Fairness, Accountability, Transparency and Human oversight [5]. However, bias may be introduced if datasets are imbalanced or if textual patterns are unrepresentative that could have disproportionate impact on certain employment sectors or organisations and it is important to assess and mitigate for bias.

In addition, privacy protection enhances the integrity of research, pre-processing and evaluation steps are reported and documented to make experiments reproducible. Implement explainable AI techniques to explain the prediction outcome – this will help the stakeholders to better understand the behaviour of the models. Furthermore, false-positive classification may lead to reputation loss, loss of confidence in the brand and employer and extra costs for the company because of wasted operations, and thus it is essential to have correct evaluation measures to ensure a balance between both sides of the equation [6].

1.10 Report Structure

This report has five chapters. Chapter 1 is the introduction to the research problem, objectives of the research, motivation, overview of research methodology, legal and ethical issues and general context of the research. Chapter 2 introduces a critical literature review of the relevant technologies, existing approaches for fraud detection and the research gap. The research methodology, dataset analysis, data preprocessing, feature engineering, development of the models and implementation process are presented in chapter 3. Chapter 4 discusses the results of the experiments using an appropriate performance measure and compares it with the previous works and models explainability is discussed. Finally, Chapter 5 synthesizes the research and provides contributions, limitations and recommendations for future research.

Chapter 2: Literature Review and Technology Review

2.1 Introduction

The literature review offers in-depth analysis on the state of the art of fraudulent job advertisements and highlights the areas that need to be explored to warrant the current study. It provides a summary of past research within the area of online recruitment fraud, Natural Language Processing, machine learning, advanced language representation techniques and explainable Artificial Intelligence, along with the effectiveness, strengths and limitations of the studies. Furthermore, the paper provides an overview of the most popular datasets, feature engineering techniques, evaluation methods and explainability techniques in the recruitment fraud detection domain. This chapter outlines the theoretical and technological background for the selection of methodologies that allow the implementation of the proposed methodology through a critical examination of the previous research and activities that support the design, implementation and evaluation of the proposed framework.

2.2 Online Recruitment Fraud

Internet recruitment scams are fake job offers set up to trick individuals into disclosing some information, paying upfront or having their identity stolen. With the quick expansion of digital recruitment platforms, scammers have been in a position to engage in recruitment scams across several platforms and social media, making them more prevalent and convincing. Today attacks are often characterized by phishing, impersonating a trusted institution, fake corporate websites and AI-generated messages, which are all able to evade traditional detection techniques [7]. These fraudulent activities lead to a considerable loss of money for people and organisations, and it erodes the confidence of online recruitment ecosystems.

In addition to the monetary damages, victims of identity thefts are also likely to experience identity theft, violations of their privacy, psychological harm and extended unemployment caused by the bogus job [8]. Recruitment fraud, from an organisational point of view, can lead to a loss of employer's brand, increase cost of compliance and require ongoing recruitment site monitoring. The issue also poses significant cyber security concerns as a fraudulent advertisement can be a gateway to stealing user credentials, malware and other social engineering attacks for sensitive data [8]. Thus, online recruitment frauds have gone from a one-off scam to a cyber-enabled threat and must be addressed through a multidisciplinary approach.

Organisations that understand the changing landscape, financial and economic consequences, and cyber security concerns surrounding recruitment fraud can create effective, transparent, and trustworthy AI-driven detection systems to protect the organisation and its employees.

2.3 Fake Job Advertisement Detection

Fake job advertisement detection has evolved from simple manually-created filters to AI-powered solutions. The initial detection techniques involved manual verification, blacklist filters and basic key words matching rules such as the verification of suspicious key words such as upfront payment and unrealistic salary. While it was easy to explain rule-based systems and they were effective in computation, they were incapable of adapting to changes in the way that language was used, and produced a lot of false positives and false negatives. Recent studies have shown that, machine learning algorithms can be more generalising by learning discriminative patterns from labelled recruitment data. Logistic Regression, Support Vector Machines, Random Forests and XGBoost are some of the algorithms which have demonstrated good performance along with text feature extraction methods like TF-IDF and n-grams [9].

To go a step further, deep learning techniques were used which automatically acquired the semantics from the textual contents to detect fake job offers. Recurrent neural network (RNN), LSTM network (LSTM), and Bidirectional LSTM network (Bilstm) better capture the contextual relationship than the traditional machine learning models, especially for the recruitment narratives with subtle deception cues [10]. These architectures typically require a lot of labels and powerful computing power, however, both of which can be difficult to acquire in a low-resource environment.

Over the past few years, a new type of architecture based on transformer is becoming the de facto standard method to detect recruitment frauds because it can model the contextual relationships using self-attention mechanism and transfer learning. These models like BERT, RoBERTa or domain specific transformer models can capture the nuances of the language patterns and can deliver more improved precision, recall and F1-scores [11]. Yet, although these developments have occurred, transformer models continue to be complex and potentially less interpretable. The combination of Explainable Artificial Intelligence and powerful language models has become an important research direction to develop a trustworthy, accurate and transparent fraud detection system for recruitment that can be applied in real-world situations. Comparative assessments also involve extensive validation, balanced performance metrics, cross validation, external testing and class imbalance analysis to guarantee reliable and generalisable assessment results every time across a variety of recruitment platforms.

2.4 NLP Feature Representation Techniques

For the development of models for fake job advertisement detection using Natural Language Processing (NLP) techniques, the choice of representation of features is very important. In the traditional models such as Bag of Words (BoW), only the frequency of words in a document will be considered, word order and word relationship are ignored, so the context knowledge will be limited [12]. TF-IDF is sparse, interpretable and assigns higher scores to informative words and lower scores to the more common words, which is appropriate for classical machine learning models [13]. Short phrases in context can be captured by using N-grams, but this is likely to add significantly to the dimension of the features to be used in discriminating deceptive expressions.

Distributed embedding, such as word2vec, GloVe, fastText and doc2vec, models create distributed dense representations with a preservation of semantic similarity. Word2Vec and GloVe are good for learning contextual word meaning, but cannot learn meanings of unknown words. The negative of this is that FastText could be used for subwords and Doc2Vec could be used for embedding representations of entire documents for classification. These however provide fixed-embeddings, which are the identical embeddings in various context of use [14]. Contextual language models, such as ELMo, BERT, RoBERTa, DistilBERT and other transformer embeddings can be used to overcome this limitation, and will provide representations which are sensitive to the context. Bidirectional attention is employed in BERT to capture complicated semantic relationships, whereas RoBERTa optimises pre-training procedures for better performance. DistilBERT reduces the computational requirement without compromising the performance; Sentence-BERT is a semantically meaningful sentence embedding suitable for similarity and classification. Although these algorithms, based on transformers, have proven to be state-of-the-art for many NLP tasks and recruitment fraud detection, they are costly, complex to compute and require vast training time and hardware expertise [15].

TF-IDF and traditional machine learning models are a suitable methodological approach due to considerations of computational efficiency, interpretability, size of the datasets and scope of the projects. TF-IDF is understandable and efficient in calculation, requires low memory resources, can be easily applied in common classification algorithms and is good for comparative evaluation thus it is suitable for using Explainable Artificial Intelligence techniques. However, TF-IDF might be able to strike a balance between the accuracy,

scalability, computational cost and interpretability of the model, even though transformer-based models might offer a greater prediction accuracy.

2.5 Machine Learning Algorithms used for Detecting Fake Job Postings

For fraudulent job postings, machine learning is the solution of choice because it can be tricky to notice the patterns in the language and contextual components of such frauds. Many studies have assessed the performance of various classification algorithms over the Employment Scam Aegean Dataset (EMSCAD) and other recruitment datasets, showing that none of the algorithms can be consistently better than the others, depending on the distribution of the data and feature engineering techniques used.

Goud and Reddy tried a variety of representations such as Bag-of-Words, TF-IDF, word embeddings, transformer embeddings, and different classifiers such as Support Vector Machine (SVM), Stochastic Gradient Descent (SGD) and Gradient Boosting. The best model was their Gradient Boosting model which has 0.88 weighted F1 score compared to their traditional lexical features. Given that the combination of handcrafted and semantic features would improve the classification performances is demonstrated. The work, however, was more geared towards fraudulent job type categorisation instead of binary fraud detection and there was not much discussion about severe dataset imbalance which could impact model generalisability [16].

Korantemaa proactively investigated the conventional and ensemble learning algorithms for the Ghanaian job website recruitment fraud detection. The best accuracy was achieved by Random Forest (91.86%) and the other feature selection techniques Information Gain and Chi-square showed little drop in accuracy (91.51%). Evaluation of accuracy and AUC were used in the study to show good results for ensemble methods, but the information regarding precision, recall and class specific error were not evaluated adequately. This evaluation strategy might overestimate effectiveness in practice since the datasets are naturally imbalanced in terms of recruitment fraud [17].

By combining the TF-IDF and metadata features, Salloum et al. got high accuracy of 96.78% in the classification. The outcome of the study showed that significant improvement has been obtained by the combination of Textual and Structured attributes, in both fraud detection capability and efficient computation. However, Logistic Regression has the disadvantage that it models linear decision boundaries which are not as expressive as other learning models in the ability to capture complex semantic relationships in deceptive recruitment language [18].

Nanath and Olney delved into hybrid methods of machine learning and crowdsourcing for online recruitment fraud detection. They discovered Decision Tree and Generalised Linear Models were the most explanatory models in the automated solution and that crowdsourced credibility signals also contributed to improve classification accuracy. The one important thing that this study noted was that human intelligence can be used to detect automated detection. On the other hand, the use of crowdsourcing also has its drawbacks including problems with scalability, higher operational costs and inconsistencies due to human subjective judgment [19].

In the same manner, Al Zahrani et al. have conducted a comparison between the deep learning-based NLP techniques, and the deep learning neural architectures have been found to be significantly better in terms of the precision, recall and F1-score as it was able to retrieve more contextual semantics. They concluded that their results supported using representation learning over hand-crafted features. But these models have higher hardware requirements, longer training periods and bigger labelled datasets, which makes them less feasible for companies with limited hardware [20].

In this regard, Taneja et al. designed a transformer-based framework named as Fraud-BERT which was evaluated on EMSCAD. Using contextual embeddings and transfer learning, the proposed model obtained around 0.99 accuracy and 0.93 F1 score which is superior compared to the previous machine learning and deep learning approaches. Although these impressive successes have been achieved, transformer models are costly for computation, not interpretable and require particular hardware. Moreover, achieving very high accuracy on one benchmark dataset brings into question the external validation (i.e. generalisability across platforms) and the requirement for extensive validation on multiple metrics and Explainable Artificial Intelligence techniques [21].

2.6 Explainable AI

With many of the most powerful models functioning as a black box, decisions made are hard to interpret, and this is why Explainable Artificial Intelligence (XAI) has become an integral part of the current Machine Learning landscape. Explainability helps to understand the dynamics that underlie model decisions, enhancing transparency, accountability, and trust for the stakeholders in the context of fraud detection. SHapley Additive exPlanations (SHAP) is a game-theory based method, that is consistent across all features, over the space of prediction and can be used for local explanations [22]. On the other hand, Local Interpretable Model-agnostic Explanations (LIME) learns a simpler surrogate model locally around each instance

to provide a detailed explanation to a particular prediction, which is irrespective of the underlying model [23].

While global explanations give an overall understanding of the model behaviour as they identify the most influential features throughout the entire dataset, local explanations help analysts validate specific predictions, for instance, whether a specific job advertisement is classified as fraudulent or not. In the context of recruitment fraud detection, these features can help ensure compliance with regulations, enhance human oversight, and minimize the need for opaque decision-making. Furthermore, Explainable AI can contribute to the principles of trustworthy AI by ensuring fairness, accountability, transparency, and responsible use, while keeping automated systems that detect fraud in recruitment processes interpretable, auditable, and suitable for their high-stakes nature, and maintaining high levels of predictivity and stakeholder trust [24].

2.7 Research Gap

A significant amount of advancement has been witnessed in the literature pertaining to fraud detection of job postings through machine learning, deep learning and transformer based Natural Language Processing models. But there are a number of key issues that are not being addressed. First, most studies focus on predictive accuracy, and do not pay attention to the transparency or interpretability of the model, making it hard for recruiters to understand why a vacancy is identified as fraudulent. Second, the majority of research is only validated with one benchmark data set, and thus, the models of other recruitment platforms and the evolution of fraud isn't widely transferable. Third, the information sets used in recruitment fraud can sometimes be imbalanced, which means that models can be biased towards the genuine postings. In addition, there are some methods based on transformers that achieve considerable prediction performance, but require heavy computation and are therefore not practical for small and medium-sized organizations.

Furthermore, previous studies have not been conducted on a proper systematic comparison of multiple conventional machine learning algorithms under the same experimental set-up, and the application of Explainable Artificial Intelligence (XAI) techniques. Therefore, a sufficient amount of evidence on the trade-off between predictive performance, computational efficiency and interpretability does not exist. The present work aims to fill this gap by designing an interpretable fake job advertisement detection framework by using TF-IDF feature representation, multiple machine learning algorithms and explainable artificial intelligence methods. The proposed method is to obtain competitive classification results, explain the

decisions made by the classifiers, efficient computation and applicability to real world recruitment fraud detection systems.

Chapter 3: Methodology and Implementation

3.1 Research Design

The type of research is quantitative research, experimental and comparative research, and framework construction was carried out to test the constructed framework using artificial intelligence to detect fake job advertisements. Here, a quantitative approach was selected since its model performance can be objectively measured using numerical evaluation metrics including accuracy, precision, recall, F1-score and Precision-Recall curve. This approach enables the statistical comparison of the different machine learning algorithms and provides reproducible and impartial results.

This was a research experiment as the system was tested with the various classification models in a controlled environment. The experimental process includes data pre-processing, feature extraction through TF-IDF method, training of model, hyperparameter optimisation and testing of the trained model on the same training and testing data sets. To ensure that any differences in the predictive performance was due to the difference in the learning algorithm rather than due to inconsistencies in data preprocessing, the same standardised preprocessing procedures were used.

The study further uses a comparative research design to evaluate the different supervised machine learning algorithms such as Naive Bayes, Linear SVM, Random Forest and XGBoost. Through comparative analysis, the most appropriate model in terms of predictive accuracy, computational efficiency, interpretability of the model and robustness of the model in dealing with imbalanced recruitment data was identified. Furthermore, the Explainable Artificial Intelligence techniques were applied to compare the importance of the features so that it helps to better understand the model. An integrated research design, that in a systematic, rigorous and reproducible way, offers an accurate, interpretable and practically deployable fake job advertisement detection system.

3.2 Development Environment

The work is implemented in Python and on Google Colab Cloud Computing platform, which provides an easy-to-use platform for developing the model without being required to set up any hardware. The software environment was set up using Python 3.x along with data manipulation libraries such as Pandas and NumPy, visualisation libraries like Matplotlib and Seaborn, text pre-processing library NLTK, TF-IDF feature extraction library Scikit-learn, model training and evaluation library Scikit-learn, class imbalance library Imbalanced-learn (SMOTE), and Explainable Artificial Intelligence libraries SHAP and LIME. The cloud computational

environment resources (Google Colab) were sufficient and provided multicore CPU and/or adequate RAM and GPU for the use of the resources, as needed. The platform enables efficient experimentation, reproducible workflows, package management, interactive visualisation and smooth machine learning workflows during research.

3.3 Dataset Description

The data used for this research is obtained from Real or Fake Job Posting Prediction dataset from Kaggle repository which was originally created by the Laboratory of Information and Communication Systems Security (LICS), University of the Aegean. This dataset can be found on Kaggle (<https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction>). It contains 17,880 job postings that were mined from online job markets, and is popular for use as a reference information set in the research of recruitment fraud detection.

There are 18 attributes of the data set, 17 of which are used to predict and the target attribute is 'fraudulent'. All the textual variables (title, company_profile, description, requirements, benefits) and the metadata (location, department, salary_range, employment_type, required_experience, required_education, industry, function, telecommuting, has_company_logo, has_questions) are the predictor variables. These variables have semantic and contextual information that can be used to analyse recruitment ads extensively.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17880 entries, 0 to 17879
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   job_id                                17880 non-null  int64
1   title                                 17880 non-null  object
2   location                              17534 non-null  object
3   department                            6333 non-null   object
4   salary_range                          2868 non-null   object
5   company_profile                       14572 non-null  object
6   description                           17879 non-null  object
7   requirements                          15184 non-null  object
8   benefits                             10668 non-null  object
9   telecommuting                        17880 non-null  int64
10  has_company_logo                      17880 non-null  int64
11  has_questions                         17880 non-null  int64
12  employment_type                       14409 non-null  object
13  required_experience                    10830 non-null  object
14  required_education                    9775 non-null   object
15  industry                              12977 non-null  object
16  function                              11425 non-null  object
17  fraudulent                            17880 non-null  int64
dtypes: int64(5), object(13)
memory usage: 2.5+ MB
```

Figure 3.1: Dataset Details

The target variable is binary, with 0 indicating legitimate job ads, and 1 indicating fraudulent job ads. There are 17014 legitimate and 866 fraudulent postings indicating that the minority class is 4.84% of observations and the class distribution is highly imbalanced with the imbalance ratio of about 19.6:1. This imbalance poses a challenge in that classification models can be biased towards the majority class, and the application of techniques such as the Synthetic Minority Oversampling Technique (SMOTE) are important in developing a classification model.

A lot of the metadata fields are also missing, such as salary_range, department, benefits, required_education and required_experience and most of the textual fields are not filled in. Therefore, data cleaning, handling the missing data, removing stop words, lemmatisation and TF-IDF feature extraction were done as training the model. Overall, all the above-mentioned aspects of public availability, realism of class imbalance, structured metadata and rich textual information make the EMSCAD dataset suitable for developing and testing machine learning models for fake job advertisement detection.

3.4 Data Preprocessing and Quality Assessment

The data was thoroughly processed to ensure high data quality, little textual noise and optimal predictive outcomes before model building. Some metadata elements had many unknown qualities in quality check #1. Figure 3.2 shows that salary_range had 83.96% of missing values, department had 64.58% of missing values, required_education had 45.33% of missing values, benefits had 40.34% of missing values and required_experience had 39.43% of missing values. These observations were not deleted, but the missing values were kept as the textual information in most of these observations was complete and the observations contained valuable data for recruitment. This method reduced loss of data, and kept the original data size (17,880 records). These are very sensitive to effectiveness of traditional machine learning models and modern NLP models, depending on suitable pre-processing [25][26].

```

=== Missing Values Percentage ===
salary_range      83.959732
department        64.580537
required_education 45.329978
benefits          40.335570
required_experience 39.429530
function          36.101790
industry          27.421700
employment_type    19.412752
company_profile    18.501119
requirements       15.078300
location          1.935123
description        0.005593
dtype: float64

=== Target Class Distribution ===
fraudulent
0      0.951566
1      0.048434
Name: proportion, dtype: float64

```

Figure 3.2: Missing Values and Target Class Distribution

The target variable assessment also revealed that the data was greatly skewed with 95.16% legitimate ads and 4.84% fraudulent ads. This imbalance may lead classifiers to be biased toward the majority class which may make the fraud detection more challenging. Next, in order to avoid discarding legitimate samples, Synthetic Minority Oversampling Technique (SMOTE) was added to the training to make learning minority class easier.

The main textual features like title, description, requirements, benefits and company_profile were cleaned by the text cleaning procedure depicted in Figure 3.3. All letters were converted to lower case for lexical consistency, URLs and HTML tags were eliminated to filter out irrelevant information on the Web, special characters and numerical symbols were eliminated to avoid noise, English stop words were eliminated to keep meaningful ones and WordNet lemmatisation was performed to reduce the noises of inflected words. These operations meant to decrease the sparsity of these features maintained the semantic information in a good TF-IDF representation [25].


```

def clean_job_text(text):
    """
    Performs comprehensive NLP preprocessing:
    1. Lowercasing
    2. Removing URLs, HTML tags, and special characters
    3. Removing stopwords
    4. Lemmatization
    """
    if not isinstance(text, str):
        return ""

    # 1. Lowercase
    text = text.lower()

    # 2. Remove URLs and HTML-like tags
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'<.*?>', '', text)

    # 3. Remove special characters and numbers (keeping only letters and spaces)
    text = re.sub(r'[^a-zA-Z\s]', '', text)

    # 4. Tokenize and remove stopwords
    stop_words = set(stopwords.words('english'))
    tokens = text.split()
    filtered_tokens = [word for word in tokens if word not in stop_words]

    # 5. Lemmatization
    lemmatizer = WordNetLemmatizer()
    lemmatized_tokens = [lemmatizer.lemmatize(word) for word in filtered_tokens]

    return ' '.join(lemmatized_tokens)

```

Figure 3.3: Function for Cleaning Text Data

All of the cleaned textual features were then assembled together as a master_text_feature (as shown in figure 3.4) to get a single representation of the document for feature extraction. Empty combined text rows were deleted but there were no observations with no meaningful content; that is, the data set was not changed, and remained at 17,880 observations.

```

# Apply cleaning to key textual columns
text_columns = ['title', 'description', 'requirements', 'benefits', 'company_profile']
for col in text_columns:
    job_postings_df[f'{col}_clean'] = job_postings_df[col].fillna('').apply(clean_job_text)

# Combine all cleaned text into one master feature for TF-IDF
job_postings_df['master_text_feature'] = job_postings_df[[f'{col}_clean' for col in text_columns]].agg(' '.join, axis=1)

# Remove rows where the master text is empty after cleaning
job_postings_df = job_postings_df[job_postings_df['master_text_feature'].str.len() > 0].reset_index(drop=True)
print(f"\nData shape after cleaning empty texts: {job_postings_df.shape}")

```

Data shape after cleaning empty texts: (17880, 25)

Figure 3.4: Applying Text Cleaning Operations

Lastly, the feature engineering operations are shown in figure 3.5. However, binary indicator variables have been created in lieu of discarding records where information about some of these variables was not provided (has_salary_info, has_department, has_education_req and has_experience_req). These variables provide some insight into whether recruitment information is available or not, which can be an indicator of fraudulent behaviour. These engineered features are employed with TF-IDF representation and, along with the binary

features mentioned above (telecommuting, has_company_logo, has_questions) they contribute to the discrimination of the models without introducing additional computation [27].

```
# FEATURE ENGINEERING
# Instead of dropping rows with missing salary/department, we create binary flags.
# Fraudulent posts often lack these details.
job_postings_df['has_salary_info'] = job_postings_df['salary_range'].notna().astype(int)
job_postings_df['has_department'] = job_postings_df['department'].notna().astype(int)
job_postings_df['has_education_req'] = job_postings_df['required_education'].notna().astype(int)
job_postings_df['has_experience_req'] = job_postings_df['required_experience'].notna().astype(int)

# Select final numerical/binary features to combine with TF-IDF later if needed
# For this NLP-focused study, we will primarily rely on the text, but these flags
# can be appended to the TF-IDF matrix for a hybrid model.
numerical_features = ['telecommuting', 'has_company_logo', 'has_questions',
                     'has_salary_info', 'has_department', 'has_education_req', 'has_experience_req']
```

Figure 3.5: Applying Feature Engineering Operations

3.5 Exploratory Data Analysis

Through the exploratory data analysis, insights were gained on the nature of the fake job posting data and this was helpful in making modelling decisions. The ratio of the target class is highly imbalanced as shown in Figure 3.6 with fewer fraudulent job postings than legitimate ones. This imbalance shows that if the classifier model is trained without proper mitigation techniques, it could have a bias towards the majority class, leading to insufficient detection of fraudulent ads. Therefore, it is concluded that SMOTE can be used in training of the model to boost up minority class which has the effect of better performance of classification.

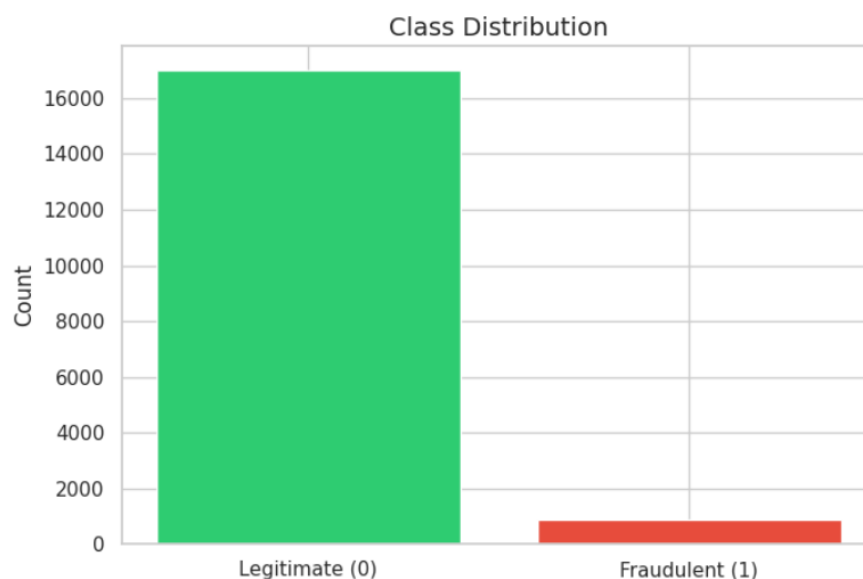


Figure 3.6: Target Class Distribution Visual

The length of the job description is shown for legitimate and fraudulent ads in figure 3.7. There's a lot more variation in the length of description in legitimate postings, and many extreme outliers, some of which are ads longer than 14,000 characters. There are some

differences in the posting that's fraud, and it's not as many super long descriptions. The two distributions have a significant overlap, indicating that description length cannot be the sole criterion to differentiate fraudulent advertisements. But it can also be assumed that the information from texts can be complemented by the lexical characteristics obtained from the TF-IDF.



Figure 3.7: Description Length by Fraud Status

In the correlation matrix shown in Figure 3.8, the correlation between selected binary variables with the class is examined. The most statistically significant correlation with fraud is seen for `has_company_logo` and this variable has a moderate negative correlation of -0.26 with fraudulent advertisements. This is an interesting discovery as it indicates that "the presence of an official company logo" could be a predictive feature as fake job postings are less likely to have one. In comparison, `telecommuting` has a very weak positive correlation with fraud (0.035), and weak negative correlation with fraud (-0.092) for `has_questions`. The results indicate that the predictive power of each of the metadata components is rather low individually, but can be utilized to improve the detection accuracy by fusing with the textual features in a machine learning framework.

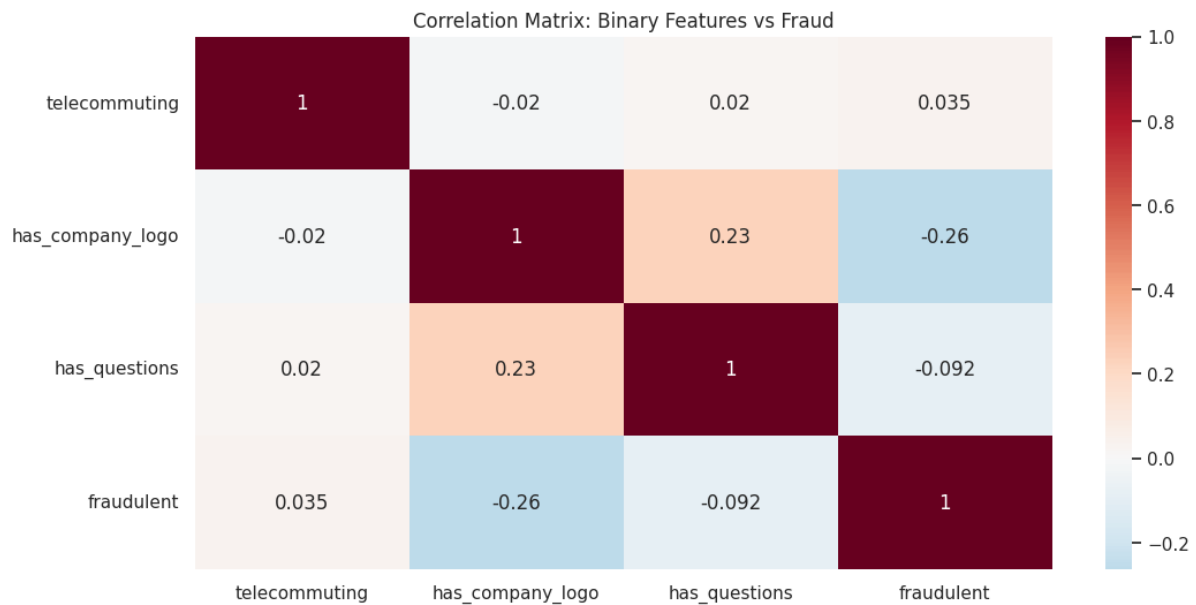


Figure 3.8: Correlation between Variables

3.6 Feature Extraction

After text processing, a machine learning friendly feature vectorization of the cleaned textual data was done using the Term Frequency–Inverse Document Frequency (TF-IDF) method. There were 15000 words in the vocabulary, which was limited through the `max_features` argument of the `TfidfVectorizer` to balance the number of features selected and the number of calculations. A range of (1,2) was chosen for the `n-gram` to capture both unigrams and meaningful word pairs (bigrams) that can help identify deceptive phrases which might be undetectable when using the unigrams alone. The sparsity of features and extremely rare terms which do not contribute much to the generalisation were eliminated by the parameter `min_df = 3`; and the terms that are too common were filtered by `max_df = 0.90`. Additionally, a sublinear (logarithmic) term frequency scaling was used, which was able to downscale the impact of highly occurring words, making the features more balanced.

```
# FEATURE EXTRACTION (TF-IDF)
tfidf_vectorizer = TfidfVectorizer(
    max_features=15000,      # Increased slightly due to better cleaning
    ngram_range=(1, 2),     # Unigrams and Bigrams
    min_df=3,               # Ignore very rare words
    max_df=0.90,            # Ignore very common words
    sublinear_tf=True       # Log normalization
)

tfidf_matrix = tfidf_vectorizer.fit_transform(job_postings_df['master_text_feature'])
feature_names_list = tfidf_vectorizer.get_feature_names_out()
```

Figure 3.9: Feature Extraction using TF-IDF

TF-IDF was chosen because: It is an efficient representation; It is interpretable; It is computation-light and it has been demonstrated to be effective with traditional supervised machine learning algorithms like Naïve Bayes, Support Vector Machine, Random Forest and Gradient Boosting. Unlike dense embedding approaches, such as Word2Vec, GloVe or transformer embedding approaches, TF-IDF is much less demanding in terms of computational resources, is much easier to implement and does not require any pre-training, and is appropriate for moderate size datasets, such as EMSCAD. In addition, representations based on transformers have been shown to have a richer understanding of the semantics, while TF-IDF has been effective in the context of document classification, and also offers greater transparency and training speed [28][29] TF-IDF also generates sparse feature matrices which can be more easily utilized by Explainable Artificial Intelligence (XAI) approaches, such as SHAP and LIME, to gain a better understanding of the most important features that influence fraud detection decisions. Therefore TF-IDF is a suitable measure of predictive performance, interpretability, scalability and efficiency for this research.

3.7 Handling Class Imbalance

After the train–test split this data set was very imbalanced with 13,611 job offers labelled as legitimate and 693 job offers labelled as fraudulent. For the first time the distribution of each class in the training set was matched with the distribution in the testing set to maintain the natural distribution of fraud. Then the data augmentation technique called Synthetic Minority Oversampling Technique (SMOTE) was used to create synthetic samples in the training data only (no information was leaked to the test data), so as to guarantee a good evaluation of the model.

The sampling method with sampling strategy 0.5 was shown in Figure 3.10 that the minority class was sampled from 693 to 6805 making the minority class 50% of the majority class. SMOTE interpolates between minority class samples that are close to each other to create synthetic minority class samples, which enables better learning of the model, and also to avoid overfitting problems that occur if using simple random oversampling [30]. The oversampling ratio used was moderate so as to enhance the minority representation without mixing up the original data distribution to make it balanced.

```

# Prepare target
y_labels = job_postings_df['fraudulent']

# Split data (Stratified to maintain imbalance ratio in test set)
X_train_tfidf, X_test_tfidf, y_train, y_test = train_test_split(
    tfidf_matrix, y_labels, test_size=0.2, random_state=42, stratify=y_labels
)

# HANDLING CLASS IMBALANCE (SMOTE)
# Apply SMOTE ONLY to training data
smote_sampler = SMOTE(random_state=42, sampling_strategy=0.5) # Oversample minority to 50% of majority
X_train_resampled, y_train_resampled = smote_sampler.fit_resample(X_train_tfidf, y_train)

print(f"\nTraining set before SMOTE: {y_train.value_counts().to_dict()}")
print(f"Training set after SMOTE: {pd.Series(y_train_resampled).value_counts().to_dict()}")

Training set before SMOTE: {0: 13611, 1: 693}
Training set after SMOTE: {0: 13611, 1: 6805}

```

Figure 3.10: Using SMOTE to Handle Class Imbalance

Random undersampling was not selected as it discards useful legitimate observations and consequently decreases the training information. Oversampling minority cases can give rise to overfitting in a similar fashion. Other extensions (Borderline-SMOTE, ADASYN) were not used as the simple SMOTE algorithm is a well-established baseline method for binary text classification tasks and has large empirical support [31]. From the above, it can be seen that SMOTE helped the classifier to classify the fraudulent ads and at the same time, helped in the representative evaluation of unseen test data.

3.8 Model Development and Evaluation Metrics

After feature extraction and class balancing as shown in Figure 3.11, four supervised machine learning algorithms have been developed and tested on the tested data: Multinomial Naïve Bayes (MNB), Linear Support Vector Machine (Linear SVM), Random Forest (RF) and Gradient Boosting (GB). They follow a variety of learning paradigms and have all proven to be effective in high dimensional text classification problems; hence they were selected.

The Multinomial Naïve Bayes was used as a baseline classifier as it is simple and has a good performance with sparse TF-IDF features [32]. However, it does not account for the relationship among words, that is, conditional independence, making it difficult to consider the relationship between complex words in the job description. Linear SVM was chosen as its high dimensional sparse representation of text data is particularly suitable and also due to the incorporation of class weights which is suitable for dealing with imbalanced sets of classes with a low overhead. Random Forest was added because it is an ensemble method which is able to model nonlinear relationships, it alleviates overfitting by bootstrap aggregation, and it offers interpretable feature importance. Gradient Boosting is chosen as a boosting based ensemble which can correct the prediction error sequentially, more representative of the

performance of ensemble classifiers than single tree classifiers and has disadvantages of more sensitive to parameters and longer training time.

```
# MODEL DEVELOPMENT & EVALUATION
models_dict = {
    "Naive_Bayes": MultinomialNB(),
    "Linear_SVM": LinearSVC(class_weight='balanced', max_iter=5000),
    "Random_Forest": RandomForestClassifier(n_estimators=150, class_weight='balanced', random_state=42, n_jobs=-1),
    "XGBoost_GB": GradientBoostingClassifier(n_estimators=150, max_depth=4, random_state=42)
}

evaluation_metrics = {}

for name, model in models_dict.items():
    print(f"\n--- Training {name} ---")
    model.fit(X_train_resampled, y_train_resampled)

    # Predictions
    y_pred = model.predict(X_test_tfidf)

    # For PR Curve, we need probabilities or decision scores
    if hasattr(model, 'predict_proba'):
        y_scores = model.predict_proba(X_test_tfidf)[:, 1]
    else:
        # For SVM, use decision function
        y_scores = model.decision_function(X_test_tfidf)
        # Normalize scores to [0, 1] for consistency in plotting
        y_scores = (y_scores - y_scores.min()) / (y_scores.max() - y_scores.min())

    # Metrics
    report = classification_report(y_test, y_pred, target_names=['Legit', 'Fraud'])
    pr_precision, pr_recall, _ = precision_recall_curve(y_test, y_scores)
    pr_auc = auc(pr_recall, pr_precision)

    evaluation_metrics[name] = {'pr_auc': pr_auc, 'model': model, 'scores': y_scores, 'preds': y_pred}

    print(f"PR-AUC: {pr_auc:.4f}")
    print(report)
```

Figure 3.11: Model Development and Evaluation

More algorithms like k-Nearest Neighbours were not considered as their computation is very costly on high dimensional TFIDF vectors. Transformers and deep learning models were also not explored as they are resource intensive, need significant hyperparameter tuning and require special equipment. The aim of this research was to compare the interpretable and computation-efficient machine learning algorithms, which can be implemented in practice in the recruitment systems.

To maintain unbiased performance estimation, for each model SMOTE-balanced training data set was used for training and untouched testing data set was used for evaluating the performance of each model. PR curves were built by getting predictions of each of the classifiers, and probability estimates/decision scores were used to obtain these predictions. The scores of the decision function were normalised between models as SVM-Linear does not provide out-of-the-box probability estimates.

Both accuracy and precision, recall, F1-score, classification reports and confusion matrices and PR-AUC were used in the evaluation of the model performance. The accuracy is an overall

accuracy of the observations that are classified correctly, and can be misleading if the data set is highly imbalanced. Precision is the percentage of the predicted fraudulent ads that are actually fraudulent ads and recall is the percentage of real fraudulent ads that are predicted. When the consequences of false positive and false negative are severe, e.g., in fraud detection, a compromise between precision and recall is a good choice: F1-score. Moreover, PR-AUC is selected because it is a better indicator of the ability of classifiers on imbalanced data sets as it prioritises detection of minority classes rather than the majority classes [33]. Confusion matrices were also used to make detailed analyses of the classification errors and detailed model comparison [34].

3.9 Explainable AI

Following the model assessment, Explainable Artificial Intelligence (XAI) techniques were implemented to interpret the predictions of the most successful model (classifier). The model with the highest PR-AUC was automatically selected to make sure that explanations were created from the most trustworthy classifier as shown in Fig. 3.12. A random sample of 30 testing instances was then selected to give representative explanations, to reduce the number of calculations performed. SHAP TreeExplainer was used for the tree-based models as it can compute exact Shapley values efficiently, and KernelExplainer was used for the non-tree-based models to maintain model agnosticism of the interpretability model. To gain a deep understanding of model behavior globally, SHAP summary plots were created to pinpoint those features that are most important in predicting fraud [35].

```
best_model_name = max(evaluation_metrics, key=lambda k: evaluation_metrics[k]['pr_auc'])
best_model = evaluation_metrics[best_model_name]['model']
print(f"\n Using Best Model for XAI: {best_model_name}")

# Sample a small subset for XAI to avoid memory/time issues
sample_size = 30
sample_indices = np.random.choice(X_test_tfidf.shape[0], sample_size, replace=False)
X_test_sample = X_test_tfidf[sample_indices]
y_test_sample = y_test.iloc[sample_indices]
original_texts_sample = job_postings_df['master_text_feature'].iloc[y_test.index[sample_indices]].values

# --- SHAP Analysis ---
# Use TreeExplainer if it's a tree-based model, otherwise KernelExplainer
if 'Forest' in best_model_name or 'Boost' in best_model_name:
    shap_explainer = shap.TreeExplainer(best_model)
else:
    # For NB or SVM, use KernelExplainer (slower)
    shap_explainer = shap.KernelExplainer(best_model.predict_proba if hasattr(best_model, 'predict_proba') else best_model.decision_function, X_test_sample[:10])

shap_values = shap_explainer.shap_values(X_test_sample)

# Plot Summary (Global Importance)
# Note: For multi-class or binary, shap_values might be a list. We take the positive class [1]
if isinstance(shap_values, list):
    shap_vals_positive = shap_values[1]
else:
    shap_vals_positive = shap_values

shap.summary_plot(shap_vals_positive, X_test_sample, feature_names=feature_names_list, max_display=15)
```

Figure 3.12: Performing Shap Analysis

Using LIME, an individual fraudulent job advertisement, as shown in Figure 3.13, was then explained. A prediction wrapper was used to first convert raw text to TF-IDF vectors before

getting the probability estimates from various classifiers. LIME perturbed the selected document and built the local surrogate model which would help identify the most influential words that are contributing to the fraud prediction. This local explanation can be used together with SHAP, and offers instance-specific interpretability. The merging of SHAP and LIME was proposed as these two come with a double-sided analysis, which would be useful for transparency, validation of the model, trust in the results between the stakeholders and compliance with the regulations, thus allowing the responsible use of AI in the field of recruitment fraud detection [36].

```
lime_explainer = lime.lime_text.LimeTextExplainer(class_names=['Legitimate', 'Fraudulent'])

# Define a robust prediction function that handles 2D array requirements
def predict_proba_for_lime(texts):
    """
    Wrapper to ensure models receive 2D sparse matrices and return probabilities.
    """
    # Transform the list of texts using the SAME vectorizer used in training
    texts_transformed = tfidf_vectorizer.transform(texts)

    if hasattr(best_model, 'predict_proba'):
        return best_model.predict_proba(texts_transformed)
    else:
        # For SVM/Linear models without predict_proba
        scores = best_model.decision_function(texts_transformed)
        # Convert decision scores to probabilities using sigmoid-like normalization
        probs = 1 / (1 + np.exp(-scores))
        # Stack to create [prob_legit, prob_fraud] format
        return np.column_stack([1 - probs, probs])

# Find a fraudulent example in the sample to explain
fraud_indices_in_sample = np.where(y_test_sample.values == 1)[0]

if len(fraud_indices_in_sample) > 0:
    idx = fraud_indices_in_sample[0]
    text_to_explain = original_texts_sample[idx]

    print(f"Explaining Job ID: {job_postings_df.iloc[y_test.index[sample_indices][idx]]['job_id']}")
    print(f"True Label: Fraudulent (1)")
```

Figure 3.13: Performing LIME Analysis

3.10 Project Management

The project was divided into distinct phases, including literature review, data preparation, model development, and model evaluation, as well as Explainable Artificial Intelligence (XAI) techniques, report writing, and final submission. At the conclusion of each phase key milestones were identified to track the progress of the project and ensure that objectives for each phase were being met. Throughout the project regular meetings were made with the supervisors in order to review the project and minimise deviations from the research plan to obtain feedback. Risks of the project were early identified (Data Quality, Model performance,

Computational constraint, time constraint etc) and measures taken to minimise impacts and ensure the project can proceed.

Table 3.1: Gantt Chart

Activities	Week 1–2	Week 3–4	Week 5–6	Week 7–8	Week 9–10	Week 11–12
Project planning	✓					
Literature review	✓	✓				
Research gap identification		✓				
Dataset collection		✓				
Data preprocessing			✓			
Exploratory data analysis			✓			
Feature extraction (TF-IDF)			✓	✓		
Model development				✓		
Model evaluation				✓	✓	
Explainable AI (SHAP & LIME)					✓	
Dissertation writing and revision					✓	✓
Final review and submission						✓

Table 3.2: Project Milestones and Risk Assessment

Milestone	Potential Risk	Mitigation Strategy
Literature review completed	Limited recent research	Use recent peer-reviewed databases (IEEE, Springer, ScienceDirect).
Dataset prepared	Missing values and class imbalance	Apply preprocessing, feature engineering, and SMOTE.
Feature engineering completed	Poor text quality	Perform comprehensive NLP cleaning and TF-IDF extraction.

Models successfully trained	Low predictive performance	Tune hyperparameters and compare multiple algorithms.
XAI implementation completed	Incompatible explanation methods	Select SHAP and LIME according to model type.
Dissertation submission	Time overruns or revision delays	Maintain weekly progress reviews and supervisor feedback sessions.

Chapter 4: Results and Evaluation

4.1 Experimental Setup

An experimental set-up has been prepared to make it fair and reproducible to test the machine learning models. The fraudulent class was considered as the target variable, class 0 represents legitimate advertisements, and class 1 represents fraudulent advertisements. The feature matrix (TFIDF) was used for training the models. Stratified train–test split was used, where 80% of the data was used for training the model, and 20% for the testing is used, and the random state is set to 42 to ensure that the class distribution is the same for both the training and testing data sets and to ensure that the results can be reproduced. The original training dataset was then used for class balancing using SMOTE and developing the models while the undisturbed testing dataset was used for only performance evaluation.

4.2 Analysis of Performance of the Models

The same set of TF-IDF features and training data set (SMOTE-balanced data set) were employed in the experiments to assess the performance of the four supervised machine learning algorithms. The performance analysis was done using the accuracy, precision, recall, F1 Score, Precision-Recall Area Under the Curve (PR-AUC), confusion matrices, SHAP and LIME (quantitative and qualitative analysis) to evaluate the effectiveness of the model in detecting fake job advertisements.

In the class imbalance scenario, the best performance was obtained by linear SVM with PR-AUC 0.9354, as presented in figure 4.1, which means that it was most effective at discriminating between legitimate and fraudulent advert in the class imbalance scenario. The model also exhibited 99% accuracy, 83% recall, 95% precision and 88% F1-score for the fraudulent class. The results indicate that Linear SVM has a very good balance for detecting fraudulent ads and reducing false alarms and is the best model to be deployed in practice.

--- Training Naive_Bayes --- PR-AUC: 0.8652					--- Training Random_Forest --- PR-AUC: 0.9231				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Legit	1.00	0.95	0.97	3403	Legit	0.98	1.00	0.99	3403
Fraud	0.51	0.91	0.65	173	Fraud	1.00	0.66	0.79	173
accuracy			0.95	3576	accuracy			0.98	3576
macro avg	0.75	0.93	0.81	3576	macro avg	0.99	0.83	0.89	3576
weighted avg	0.97	0.95	0.96	3576	weighted avg	0.98	0.98	0.98	3576
--- Training Linear_SVM --- PR-AUC: 0.9354					--- Training XGBoost_GB --- PR-AUC: 0.8633				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Legit	0.99	1.00	0.99	3403	Legit	0.99	1.00	0.99	3403
Fraud	0.95	0.83	0.88	173	Fraud	0.90	0.71	0.80	173
accuracy			0.99	3576	accuracy			0.98	3576
macro avg	0.97	0.91	0.94	3576	macro avg	0.94	0.85	0.89	3576
weighted avg	0.99	0.99	0.99	3576	weighted avg	0.98	0.98	0.98	3576

Figure 4.1: Performance of the Models

Overall, the second-best PR-AUC was obtained by the Random Forest model which had 98% accuracy. Notably it was able to achieve 100% accuracy, meaning that it was able to classify 100% of the ads that were labelled as fraudulent as fraudulent. However, 66% of fraud ads were recalled, meaning that 34% of fraud ads were marked as legitimate ads. If the need is there, then not being falsely accused is a conservative approach, but there could be a lot of fraudulent ads that don't get caught.

The PR-AUC of the Multinomial Naïve Bayes was 0.8652 and the overall accuracy was 95%. While the model got the most recall (91%) recalling most of the fraudulent ads, the precision was only 51% which in turn shows that there are many False Positive predictions. But it would involve a vast amount of manual checking in real-life recruitment situations, which will be effective at fraud detection.

Similarly, Gradient Boosting (XGBoost_GB) achieved a PR-AUC of 0.8633, 98% accuracy, 90% precision, 71% recall and an F1-score of 80%. The model obtained high precision, but low recall, indicating that it was not as successful as the other models in the detection of fraudulent ads to that of Linear SVM.

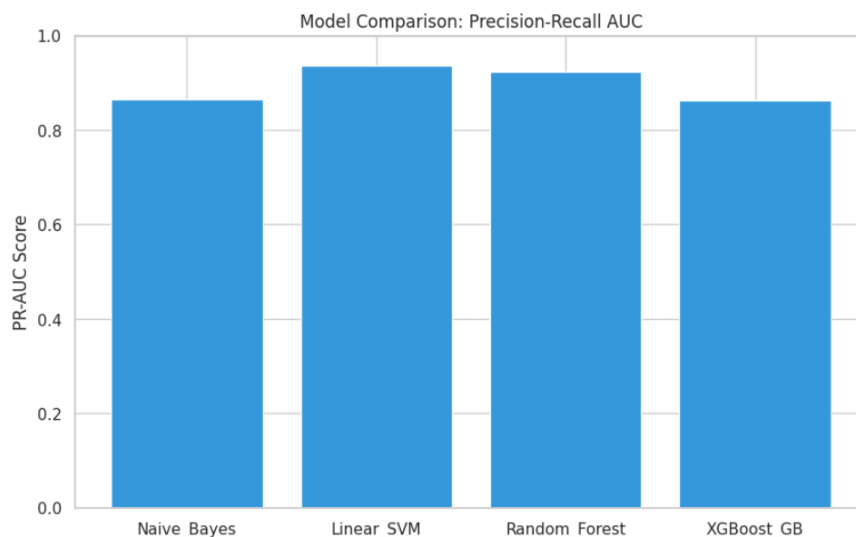


Figure 4.2: Model Comparison in terms of Precision-Recall AUC

Further information on the classification behaviour can be gained from the confusion matrices of Figure 4.3. Linear SVM had the best balance with 3,395 correctly classified legitimate and fraudulent advertisements and 8 and 30 false positives and false negatives, respectively. Random Forest has been able to achieve 100% recall without any false positives, while Naïve Bayes had 154 false positives and achieved a 100% recall on 158 out of 173 fraudulent ads.

Although Gradient Boosting managed to get 123 fraudulent advertisements correct, it misclassified 50 advertisements, with a lower sensitivity as compared to Linear SVM.

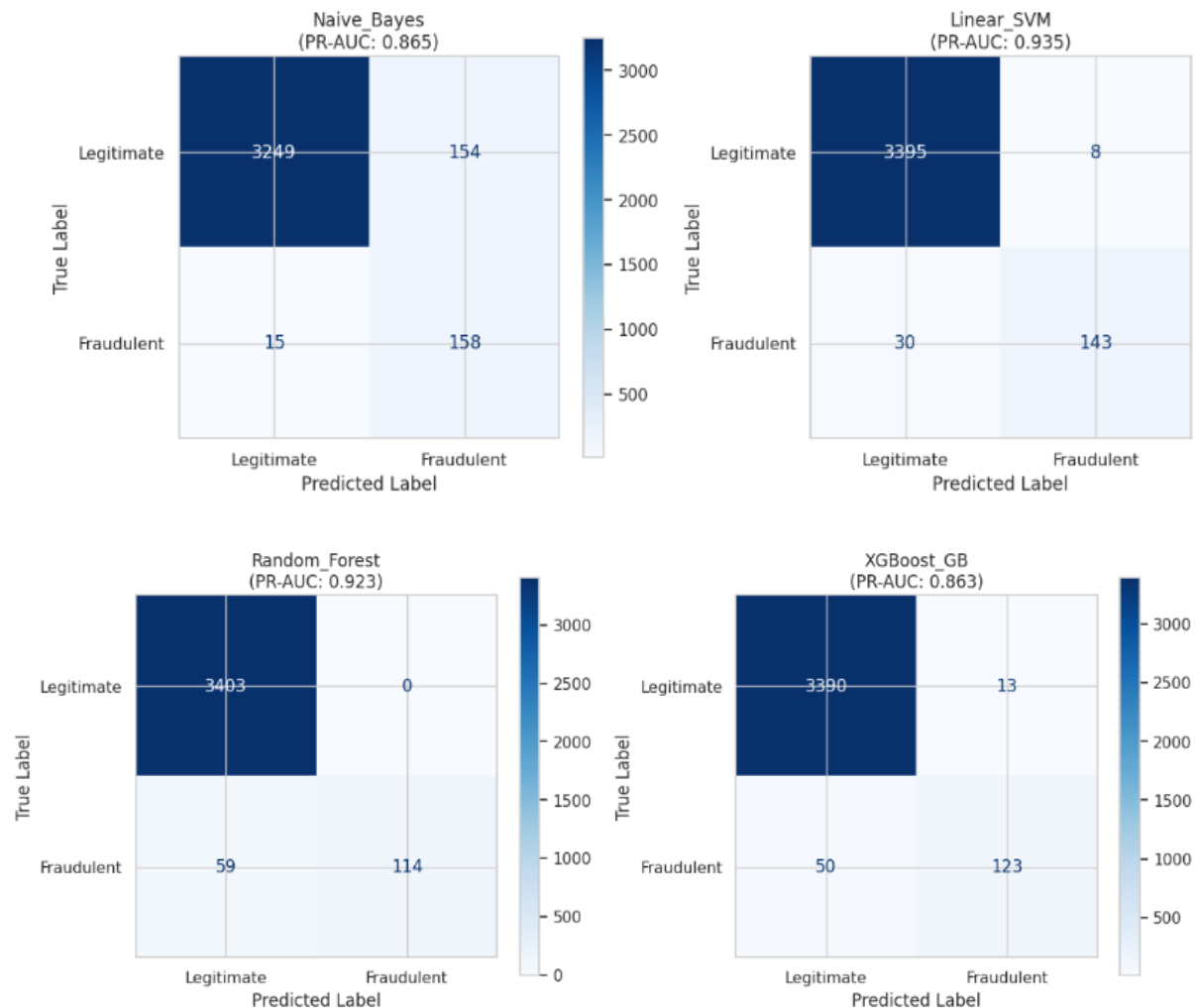


Figure 4.3: Confusion Matrix

The behaviour of the model is further validated with the results obtained from the Explainable Artificial Intelligence. The model top words that are identified as the most influential are shown in the SHAP summary plot (Figure 4.4) and include the word peoplesoft, high school, client, digital, software, marketing and signing bonus which stands for relative contribution in fraud prediction.

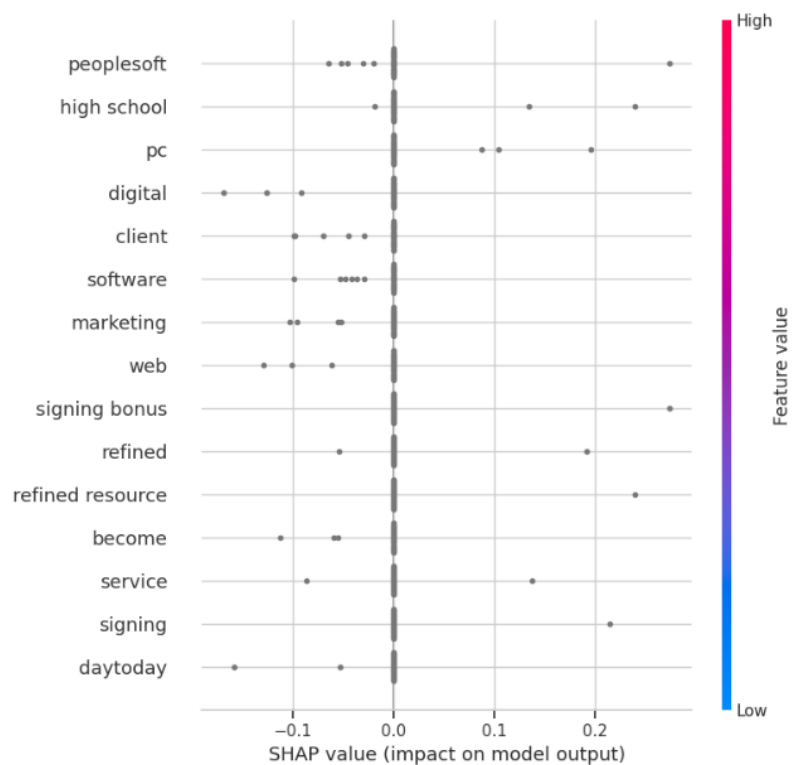


Figure 4.4: Feature Importance using Shap

In addition to this worldwide interpretation, the LIME explanation (Figure 4.5) shows that words like "refined, signing, service, required, encouraged, bonus" had a positive association with fraud classification, while words like "client, procurement" had a negative association. The explanations reveal that the classifier is learning semantically rich language patterns, rather than just coincidences.

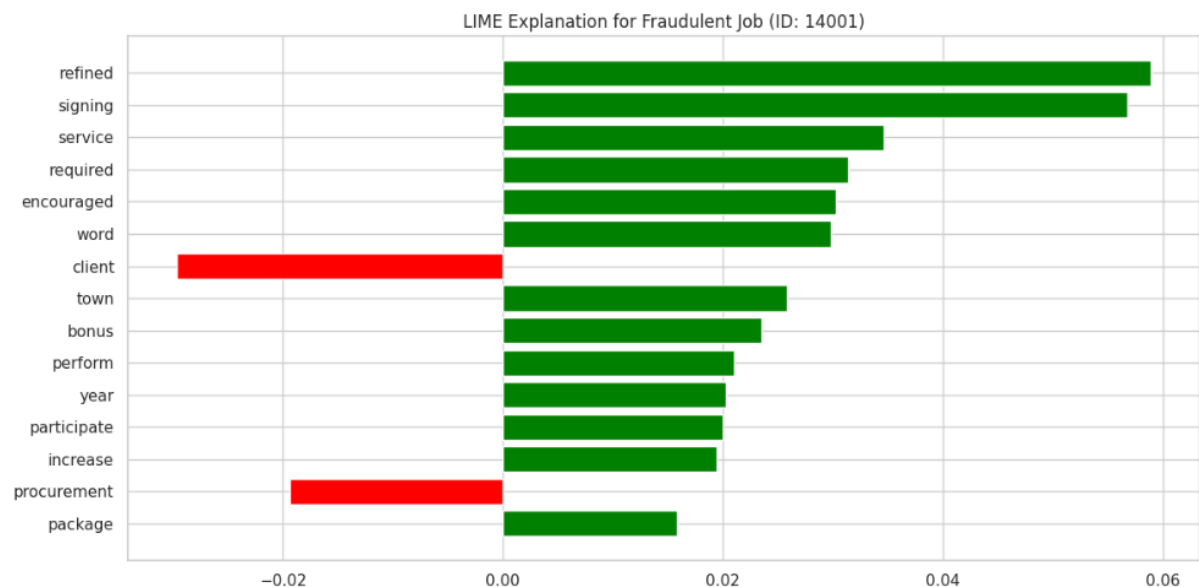


Figure 4.5: Top Features Contributing to Fraud using LIME

Lastly, the outcomes reveal that the framework TF-IDF, SMOTE and Linear SVM is the most robust, interpretable and best and most accurate framework in the fake job advertisement detection.

4.3 Comparison with Previous Research

The results of the current study corroborate with the recent studies that proved the usefulness of ML in detecting fake job postings and some discrepancies are observed in the type of models, evaluation process and the interpretability of the model. The proposed Linear Support Vector Machine (SVM) achieved the best score among the other machine learning models used in this work which was PR-AUC (0.9354), Accuracy (99%), Precision (95%), Recall (83%) and F1-Score (88%). These results validate the feasibility of the strong effectiveness of Linear SVM in case of high dimensional TF-IDF text representation.

The results are similar to the study by Goud and Reddy where they get a weighted F1 score of 0.88 with multiple feature representations with Gradient Boosting [16]. The classification performance was the same, but SMOTE, PR-AUC evaluation, and Explainable Artificial Intelligence (XAI), were applied, which provided a more complete evaluation. Likewise, the current random forest model outperformed the random forest [17] with an accuracy of 98% as compared to 91.86% respectively, showcasing a perfect precision and recall of 100% and 100% respectively, which clearly demonstrated the effect of optimisation using TF-IDF and class balancing.

The proposed Linear SVM also obtained better accuracy than one of Salloum et al. [18] that obtained 96.78% accuracy using Logistic Regression at a price of 11.8% precision and 13.5% recall. The same goes for the hybrid crowdsourcing system developed by Nanath and Olney, which employed human verification and enhanced the detection performance [19]. However, the same results can be achieved with a completely automated pipeline, as demonstrated in the current study, which will improve the scalability and lower the operational cost.

The accuracy of the transformers and deep learning models, such as Fraud-BERT [20] and Taneja et al. [21] was higher at about 99% and F1 score at 0.93 respectively. The models are however, very computationally intensive and difficult to interpret. This proposed framework combines the advantages of SHAP and LIME while providing competitive prediction results and explainability. Thus, this study provides a full solution which is not only efficient in terms of computation but also demonstrates a good classification result, and is explainable, which is more appropriate in the practical recruitment fraud detection where deployment cost and explainability are taken into consideration and desired.

4.4 Critical Discussion

The results show that the proposed framework can be an effective solution and is easily understandable in fake job advertisement detection. The use of extensive text preprocessing, TF-IDF feature extraction, class balancing with SMOTE, comparative analysis of several machine learning models, and Explainable Artificial Intelligence techniques are the strengths of the study. When comparing the various classifiers, from the results of the comparative analysis, it can be seen that the most suitable one is the Linear SVM because it not only has better Precision-Recall AUC, but also has a good balance between precision and recall. Further, the model was made transparent with the use of SHAP and LIME which provided a global and local explanation, increasing user trust and confidence to take trustworthy decisions.

But there are some drawbacks that should be acknowledged. The features employed for the study were TF-IDF features which does not fully capture the semantics of the context and also is based on term importance which does not capture the long-range dependency in language. This means that transformer LMs may not pick up on subtle deceptive patterns in the writing, quite like the human eyes would. In addition, although SMOTE was indeed an useful technique to tackle class imbalance, synthetic oversampling may not accurately reflect the actual types of fraudulent ads and thus impact model behaviour.

There are certain limitations of the study. The EMSCAD dataset, a single benchmark dataset – was used for model development and evaluation. There are different practices in the recruitment of people from all industries, countries and online recruitment agencies and they have different ways of writing and fraud. Therefore, if it is not validated further the results may not be completely applicable to unseen data sets. Moreover, only traditional machine learning algorithms were considered, and large language models and advanced transformer models were not considered due to the computational limitations and focus on offering explainability.

The resulting system can be used by recruitment platforms, HR departments, and cybersecurity teams to detect and filter out suspicious job postings, minimizing financial losses and safeguarding job seekers against recruitment scams. Furthermore, both SHAP and LIME provide explainability as they explicitly show which features have the biggest impact on each prediction, making it easier to ensure regulatory compliance and human verification. In summary, it is an extensive combination of predictive power, user-friendliness, technical transparency, and practical application that is designed for actual recruitment fraud detection situations.

Chapter 5: Conclusion

5.1 Overall Conclusion

This research successfully designed and tested the Artificial Intelligence framework to detect fake job ads using Natural Language Processing (NLP), machine learning and Explainable Artificial Intelligence (XAI). The results indicate that the proposed methodology could be used for an accurate, transparent and efficient solution to recruitment fraud detection problem.

RQ1

The investigation showed that the separation of fraudulent job offers from the legitimate ones by the application of Natural Language Processing techniques is effective. Extensive text preprocessing and TF-IDF feature extraction were done and machine learning models were found to be effective in accurate prediction of classification on highly imbalanced data set.

RQ2

Linear Support Vector Machine (SVM) was the best performing algorithm amongst all the evaluated algorithms with the highest Precision-Recall Area Under the Curve (PR-AUC) along with excellent Accuracy, Precision, Recall and F1-score. These results have proved that the linear SVM has been capable of developing the most balanced and reliable fake job advertisement detection classifier.

RQ3

The results of the Explainable AI analyses showed that several linguistic and contextual clues were detectable that were correlated with fraudulent ads. Words that were influential across the globe were displayed by SHAP, while LIME showed that words such as signing, bonus, required, service and refined were important terms which helped in predicting fraud. The lack of company information, organisational details and other facets of metadata also were a helps to effective classification.

RQ4

SHAP and LIME greatly enhanced the transparency of the models, by providing global explanations and local explanations of the models' predictions. All of these approaches helped to make the importance of features clearly understood, helped to build trust in the stakeholders, helped to do human verification and helped to ensure responsible use of AI in recruitment.

Finally, the proposed system demonstrates that TF-IDF, SMOTE, Linear SVM and Explainable Artificial Intelligence is a feasible, comprehensible and useful method for identifying fraudulent online job advertisements.

5.2 Achievement of Objectives

The objective of this study is to systematically develop, evaluate and interpret a machine learning based fake job advertisement detection framework which will help to achieve the research objectives as mentioned in the beginning of the research. Table 5.1 presents the results of each objective and the evidence that was collected in the experimental investigation.

Table 5.1: Achievement of Research Objectives

<i>Research Objective</i>	<i>Status</i>	<i>Evidence</i>
To preprocess and transform job advertisement text using Natural Language Processing (NLP) techniques for machine learning analysis.	Achieved	Text data were cleaned through lowercasing, URL and HTML removal, stop-word elimination, lemmatisation, and TF-IDF feature extraction. Feature engineering and data quality assessment produced a structured feature matrix suitable for model development.
To construct and compare multiple machine learning models to detect fake job advertisements.	Achieved	Four supervised learning algorithms, Multinomial Naïve Bayes, Linear SVM, Random Forest, and Gradient Boosting, were developed and evaluated under identical experimental conditions using the same TF-IDF features and SMOTE-balanced training dataset.
To evaluate model performance using accuracy, precision, recall, F1-score, and ROC/PR-AUC.	Achieved	Comprehensive performance evaluation was performed using accuracy, precision, recall, F1-score, confusion matrices, and PR-AUC. Linear SVM achieved the best overall performance with 99% accuracy, 95% precision, 83% recall, 88% F1-score, and 0.9354 PR-AUC.
To apply Explainable Artificial Intelligence (XAI) methods to identify important indicators influencing fraud prediction.	Achieved	SHAP provided global feature importance, while LIME generated local explanations for individual predictions. These methods identified influential linguistic features and improved model transparency,

		interpretability, and trustworthiness for recruitment fraud detection.
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In this dissertation, Fake Job Advertisements Detection framework using Artificial Intelligence based on Natural Language Processing, machine learning and Explainable Artificial Intelligence is developed. The study examined some of the latest literature on the detection of recruitment fraud, and the implementation of text pre-processing, TF-IDF feature extraction, SMOTE to handle a class imbalance, and four supervised machine learning algorithms. In the comparative evaluation, the Linear Support Vector Machine was the best performing model with high predictive performance among the different models and also has less computational time. SHAP and LIME gave insights on the transparency of the model by providing the interpretation of the influential features in fraud predictions. The proposed framework is a representation of how a system to detect recruitment fraud can be accurate, interpretable, practical, and can be used to facilitate safer online recruitment and recruitment decision making.

5.3 Research Contributions

The contribution of this research work in the field of recruitment fraud detection is meaningful academic, technical and industrial contributions using Natural Language Processing (NLP), machine learning and Explainable Artificial Intelligence (XAI). A scientific significance of the study is that from scientific perspective, it will serve as a stepping stone for the state of the art in this area in terms of having multiple supervised machine learning algorithms combined in one experiment and having a comparative study of them. Furthermore, it introduces a novel use of SHAP and LIME – to enhance the transparency and interpretability of AI models in the field of recruitment fraud detection, where several existing studies do not consider this factor. The entire research pipeline has been built, comprising of data pre-processing, TF-IDF feature extraction, class balancing using SMOTE, creation of the machine learning model, testing of the model and explainability analysis. Linear Support Vector Machine is the best performance which indicates that it is the best combination of model performance, model efficiency and model interpretability. The framework is reproducible and it can be used for future research as reference implementation.

Industrial viewpoints find the proposed framework to be applicable to online recruitment platforms, employers and cybersecurity practitioners, which will be helpful in early identification of fraudulent job posts. The use of XAI can assist in making decisions

transparently, having them verified by human intervention and increase the level of trust in the individual and in the organisation; it can also help in reducing the financial damage, reputation loss and identity theft from recruitment scams.

5.4 Limitations

The proposed framework demonstrated a satisfactory performance in the prediction of the values but there were some limitations to be noted. The research was first carried out by the use of the Employment Scam Aegean Dataset (EMSCAD), which is a single benchmark dataset. As a result, the models created in this project may not apply to other countries, industries, languages or online recruitment service providers, due to the varying styles of writing used in ads and fraud patterns that exist.

Secondly, the TF-IDF approach was used to represent Features. TF-IDF is easy to interpret, but it fails to capture longer-range or contextual language relationships like do transformer-based language models like BERT and RoBERTa. So, there may be some ways of misleading language that are really light and are not detected.

Third, synthetic samples created might not be a perfect example of real fraudulent ads due to SMOTE's benefit to minority class learning. This may affect the way the model behaves and introduce minor biases in training the model.

Moreover, only four conventional supervised machine learning algorithms were tried out. Due to time constraints and the focus on model interpretability, other advanced methods, such as transformer models, ensemble stacking models, and large language models were not explored in this study.

Finally, the given framework only included the selected set of metadata features and the text feature. In addition, external data such as recruiter reputation, company verification records and website credibility and temporal behavioural patterns were not considered, which may further boost the accuracy of fraud detection in real-world scenarios within the recruitment domain.

5.5 Future Work

Future study is desired on the exploration of context-aware transformer models namely BERT, RoBERTa models, which learns the contextual word representation and thus improves the capability of capturing the semantic relationships as compared to the TF-IDF model. More recently, it has been shown that transformer models are able to perform better than more traditional feature engineering models in the domain of recruitment fraud detection, which captures more complex language patterns [37] [38]. In the future, performance of fine-tuning

domain-specific transformer models on recruitment data sets might be further improved to increase classification accuracy and robustness.

One of the methods is utilizing Large Language Models (LLMs) for fraud detection. In addition to classification, LLMs can also provide natural-language explanations, summarise suspicious ads and help human moderators to review high-risk ads. The combined LLM reasoning and supervised classifiers can deliver a better predictiveness and interpretability at minimal manual screening time [39].

Future research also should be undertaken for multimodal learning combining the textual information with the company logos, website features, recruiters' profiles, e-mail addresses or domain names and behavioural metadata. Combining more than one information source may be beneficial in the detection of more advanced fraud campaigns that cannot be detected from text-based information [40].

In the future, there should be real time detection systems which can evaluate an ad from a deployment perspective when it is submitted to an online recruitment site. Streaming Analytics frameworks may constantly examine new ads introduced and instantly notify users about potential fraud, reducing users' exposure to malicious content.

Further, privacy-preserving strategies, like federated learning, could also be explored, allowing for the building of fraud detection models on many recruitment platforms without sharing data of their users or organisations. Additionally, Graph Neural Networks (GNNs) can be used to model the relationships between recruiters, companies, websites, contact information and job postings to find co-ordinated fraud networks as opposed to individual fake job postings.

5.6 Reflection

This research gave me opportunities to enhance my technical and research skills all the way through the dissertation. The project enabled me to experience hands-on implementation of the Natural Language Processing techniques and machine learning algorithms and Explainable Artificial Intelligence methods in the real-world Cyber-security problem. Moreover, I got to learn Data Preprocessing, Feature Engineering, how to deal with Imbalanced Classes, Model Evaluation, Model Interpretation using SHAP & LIME.

Project management skills were particularly developed with regard to careful planning, monitoring of milestones and good time management to ensure all research was completed in time. A continuous process of planning and prioritizing of literature review, experimentation, analysis and writing of a report was needed.

The research process also helped to enhance my critical thinking, academic writing and problem-solving abilities. I learned to critically review previous experiments and justify the procedures used and objectively discuss the results of the experiment. This project has been a fantastic confidence builder in the research and application of AI techniques to real world problems in the recruitment fraud detection space.

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Appendices

Appendix A: Proposal

AI-BASED DETECTION OF FAKE JOB ADVERTISEMENTS USING NATURAL LANGUAGE PROCESSING

1. Introduction

In recent years, the success and swift expansion of online recruiting services have revolutionized the recruitment process, allowing companies and applicants to connect effortlessly as well as throughout geographical ranges. But, in parallel with this digital revolution, job offers that can lead applicants to share critical personal information, financial transactions or fraudulent activity are growing, and at a record pace. As online scams evolve, there's now greater worry than ever that it will be more difficult for the public and job posting websites to determine a good job from a bad job.

The concept of AI fake job posting detection has been selected since the scams are rapidly increasing and the conventional method of job offer authentication is lengthy. The traditional systems, such as human moderation, struggle to identify intricate language patterns used by fraudsters. Recently, many research works have been indicated that *“Machine Learning”* and *“Natural Language Processing (NLP)”* techniques can be effectively applied to textual content analysis and detecting deceptive characteristics in job advertisements. For instance, in the domain of *“fraudulent job postings”*, Pillai (2023) discovered that deep learning models were able to effectively predict the fraud with high accuracy, showcasing the potential of automated systems (Pillai, 2023). Likewise, Taneja et al. (2025) proposed a *“transformer-based approach”* for the effective detection of scam job posts that considered the context-specific representation of language and tackled the issue of imbalanced data sets (Taneja et al. 2025).

It is important because recruitment fraud continues to be a threat to job seekers on the financial, psychological and privacy levels, and job seekers have lost trust in online recruitment sites. The industry has demanded for explainable and intelligent fraud detection methodologies that can make a correct prediction and a transparent decision-making process. Therefore, this work suggests the implementation of an explainable AI system using *“NLP and machine learning”* techniques for enhancing the detection of fraudulent job ads and enabling a safer online recruitment environment.

2. Problem Statement

With more organizations utilizing online recruitment platforms, the way organizations can post open positions and reach out to candidates around the world has revolutionised. Although these

Table of Contents	
1. Introduction	3
2. Problem Statement.....	3
3. Aim and Objectives	5
3.1 Aim	5
3.2 Objectives	5
3.3 Research Questions.....	5
3.4 Discussion.....	5
4. Legal, Social, Ethical and Professional Considerations	6
5. Background.....	7
5.1 Introduction to the Research Area	7
5.2 Current State of the Art.....	8
5.3 Relationship Between the Proposed Project and Existing Research	9
5.4 Establishment of the Research Area	9
5.5 Techniques and Theories Supporting the Research.....	10
5.6 Industry Relevance and Potential Impact	10
5.7 Summary	10
Reference List.....	12

giving out personal information, making payments and/or engaging in illegal activity. The size and sophistication of fraudulent job postings have also increased in tandem with the growth of online recruiting and it is increasingly difficult to spot.

The main issue that this research is focusing on is the inaccurate and inefficient detection of fake job posting on the online job recruitment platforms. The traditional means to verify the information is mostly manual or simple rule-based filter, and has limitations to detect the complex linguistic structure and also to identify the deceptive information (Anbarasu et al. 2024). Often phishing ads are formatted and written as if they were legitimate ads, making traditional detection methods less effective. The application of advanced *“Natural Language Processing (NLP) and machine learning techniques”* has been shown to improve the performance of fraud detection by analysing the contextual and linguistic characteristics in the job advertisements (Taneja et al. 2025).

There are a variety of stakeholder groups to consider in this issue. Immediate victims are the job-seekers who will lose out on money, identity theft and privacy issues, as well as emotional stress and loss of trust in the online recruitment service. The employers and recruitment agencies are affected too, as a bad reputation for the recruitment providers and the process of employment in general is lost and candidates lose trust in the process. Moreover, online job platforms are also faced with operational and legal issues while monitoring, removing and preventing fraudulent content (Swetha et al. 2023). The general labour market may be affected as well, as recruitment fraud diminishes transparency and trust in digital recruitment ecosystems.

This problem is of great interest for Academic Research as well as for Industry practice. From a business perspective, having effective fraud detection systems can help improve the security of the platforms, protect users, and build trust in online recruitment services. *“Recent studies”* have shown that the use of transformer models and deep learning techniques can greatly enhance the detection of *“fraudulent job postings”* by leveraging the intricate semantic and contextual details embedded in textual data (Vu et al. 2025; Taneja et al. 2025). Moreover, the comparative study has revealed the effectiveness of the new powerful machine learning and transformer technologies to solve a problem like recruitment fraud detection, where one class may be much more common than the other.

postings and provide insights into the linguistic and contextual clues associated with these job postings.

3. Aim and Objectives

3.1 Aim

The primary aim of this work is to create and evaluate an “*explainable Artificial Intelligence (XAI)*” based fraudulent job posting detection model in online recruitment platforms (marketplaces) based on “*Natural Language Processing (NLP) and machine learning techniques*”.

3.2 Objectives

- To preprocess and transform job advertisement text using “*Natural Language Processing*” techniques for machine learning analysis.
- To construct and contrast “*multiple machine learning*” models to detect fraud job ads.
- To measure the performance of a model using “*accuracy, precision, recall, F1 score and ROC-AUC*”.
- To use the methods of Explainable AI to discover important indicators that impact fraud decision making.

3.3 Research Questions

RQ1

- How effectively can Natural Language Processing techniques distinguish fraudulent job advertisements from legitimate postings?

RQ2

- Which machine learning algorithm has the best performance in the fake job detection?

RQ3

- What linguistic and contextual characteristics are most associated with fraudulent job advertisements?

RQ4

- How can “*Explainable AI*” improve the transparency and interpretability of fraud detection models?

3.4 Discussion

One of the biggest challenges which online job recruitment sites are facing is the existence of

Algorithmic bias is a very serious ethical issue. If the data that a machine learning algorithm is trained on is biased, then the resulting algorithm might (and does) learn patterns that negatively impact certain organizations, industries, geographical regions, or jobs. Fairness, transparency and bias-minimisation for AI systems are key factors in recent studies on the requirements for Trustworthy AI systems (Ferrara, 2023). Inaccurate predictions by AI systems can result in false negative or false positive classifications of ads as fraudulent, which can have implications for a company’s reputation and performance.

The next question of ethics is in the area of explainability / transparency. Some of the complex machine learning models are called “black box” models as it is not easy for the user to understand the decision-making process of the model. A recent study (Cheong et al. 2024) indicates that research on transparency and accountability increases trust in the usage of AI systems for decision-making. The proposed study will use Explainable AI techniques, like SHAP and LIME, to overcome this challenge by providing users with insights into the factors affecting fraud predictions.

Social factors also play a role as a faulty fraud detection system could have an impact on recruitment websites, employers and seekers of work. There is potential for fraudulent ads to be filtered as ‘false negative’ (not detected at all) and for legitimate job postings to be filtered as potentially fraudulent (Radanliev, 2025). Therefore, multiple metrics other than the accuracy should be used to analyse the performance of the model.

The study will be carried out following the principles of Responsible AI development, Fairness, Accountability, Transparency and Reproducibility (from a professional perspective). All the data pre-processing, model construction, evaluation procedures and explainability results will be documented during the project. Ethical principles in AI emphasize the need for “*transparency, accountability, and human oversight*” when implementing AI systems for fraud detection. In this, technical performance and ethical integrity will be stressed in the proposed research in developing trustworthy recruitment fraud detection solutions.

5. Background

5.1 Introduction to the Research Area

Digital recruitment platforms like LinkedIn, Indeed, Glassdoor and other online recruitment portals have revolutionized the recruitment process in terms of its accessibility and cost, as they have grown quickly. But while the online recruitment boom has also brought more job-seeking scams to trick would-be workers. These ads might appear to be valid and could be employed

distress”. As a result, there is a need for intelligent and automated solutions that are able to accurately detect fraudulent ads, and give transparent explanations for their predictions.

To solve this issue, this study proposes an explainable artificial intelligence (XAI) framework to combine “*Natural Language Processing (NLP), machine learning and XAI*”. The study will investigate whether use of textual and contextual information from job postings can be effectively used to differentiate job postings from fraudulent postings.

The “*Real or Fake Job Posting Prediction*” dataset is obtained from Kaggle, which will be utilized in answering the research questions. This dataset includes text and structured features of job postings. Data will be pre-processed, where handling missing values, normalizing texts, tokenizing texts, stop words removal and feature extraction (TF-IDF vectorization) will be performed. “*Exploratory Data Analysis (EDA)*” will be performed to discover any patterns and features of fraudulent and legitimate job postings.

Various machine learning techniques such as Naïve Bayes, Support Vector Machine (SVM), Random Forest and XGBoost will be built and compared. These models have been chosen because they have been widely used in the fields of text classification and fraud detection. Performance will be evaluated “*using accuracy, precision, recall, F1 Score and ROC-AUC*” and the best approach will be determined.

The study will be quantitative and experimental type study. This is understandable as they can have tangible measures of performance and how the machine learning models are behaving. Additionally, Explainable AI methods like SHAP and LIME will be used to gain insights into the model’s choices and uncover significant textual trends for fraud detection. Hopefully the results will contribute towards more safe and more reliable online recruitment systems and furthering the use of NLP for fraud detection.

4. Legal, Social, Ethical and Professional Considerations

The proposed research work is to develop the Artificial Intelligence (AI) system for detecting fraudulent job postings with the help of “*Natural Language Processing (NLP) and machine learning*”. The study is based on an open project available in Kaggle however some legal, social, ethical and professional issues have to be considered throughout the project.

Legally, adherence to data protection and privacy laws is a must (Hussain and Hussain, 2025). The data in the data set will be used responsibly for academic research – jobs posted and organisation details. In the pre-processing phase personal or sensitive data expressed in the transformation of the recruitment industry to online, there’s a renewed look into the research and industry for automated fraud detection solutions.

Artificial Intelligence (AI), Natural Language Processing (NLP) and machine learning technologies have proven to be potential solutions to fraud identification on the online space. In recent years these technologies have been used for recognition of fake text patterns, classification of suspicious advertisements, and enhancement of the security of online platforms. Thus, fake job advertisement detection is a growing interdisciplinary research field, which is related to machine learning, cybersecurity, information systems and recruitment technology.

5.2 Current State of the Art

In the current state of the art, fake job advertisement detection mainly relies on machine learning and deep learning techniques. Traditional classification algorithms like “*Logistic Regression, Naïve Bayes, Support Vector Machines (SVM), and Random Forests*” were used in early studies. These methods were seen to be effective in having textual and contextual features of the job advertisement to distinguish the fake ads from vacancies.

To achieve a significant improvement in this aspect, the technique of Deep learning was used. Pillai (2023) proposed a “*Bidirectional Long Short-Term Memory (Bi-LSTM)*” model for fake job detection and obtained high predictive performance by using the textual and numerical elements in it. The study found that the deep learning techniques were better in capturing the intricate relationships that would be captured using traditional machine learning techniques (Pillai, 2023).

The newer studies have taken the transformer point of view. Taneja et al. (2025) suggested a “*context-aware Bidirectional Encoder Representations from Transformers (BERT)*” framework for identifying recruitment frauds, called Fraud-BERT. The study achieved better results than conventional machine learning approach and had the prospect of transfer learning for imbalanced class fraud detection applications. The study also showed the superiority of the proposed model against the traditional machine learning techniques and revealed the role of the transfer learning in the fraud detection problem, which can be utilized in the highly imbalanced classes.

Likewise, Vu et al. (2025) suggested a deep learning technique which combined state-of-the-art techniques in the text feature extraction and deep neural network for enhanced fake job description detection. Results showed that there was a significant improvement in the

classification process, and the results confirmed the potential of the fraud detection systems to be used with NLP.

The recent comparative study also showed that the use of transformer architecture, ensemble learning approach, and advanced techniques of feature engineering are effective in detecting online recruitment fraud. These studies have shown that the performance of transformer-based methods has been consistent and has shown better results in classification accuracy and fraud detection capabilities as compared to a variety of traditional machine learning techniques.

5.3 Relationship Between the Proposed Project and Existing Research

This proposed project is an extension of the current research. Past research has proven the ability of machine learning and deep learning to successfully detect fraudulent job ads (Lerma, 2022). But there are a few disadvantages.

First, there are many studies whose focus is only the accuracy of prediction of the model, but not on the explanations of the decisions made by the model. Secondly, most studies focus on how to identify fake ads, instead of on the specific types of linguistic patterns that affect model predictions (Goud and Reddy, 2025). Thirdly, many of the advanced models are a "black-box" approach and lack transparency and thus lack trust of the user.

To overcome these limitations, the proposed study will use a combination of machine learning and "Explainable Artificial Intelligence (XAI). More specifically, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations)" methods will be used to explain classification decisions, and to find important fraud indicators. This is an extension of earlier work which focused on prediction but not interpretability.

5.4 Establishment of the Research Area

Fraud detection in the general field of machine learning has been already established and used in banking, insurance, e-commerce, cyber security and social media fraud detection. Furthermore, text-based classification research using NLP is a mature and well-established discipline which has been developed over decades both theoretically and practically (Naudé et al. 2023).

However, fake job advertisement detection is still a specialized and new research field. Although there are an increasing number of studies in this field in the past few years, it is still

present research is to build on the earlier studies and fuse the classical machine learning techniques with Explainable AI techniques to create an efficient and explainable fraud detection system. The project tackles a problem in industry with social and economic implications that span broadly and supports the establishment of a new, emerging research area.

5.5 Techniques and Theories Supporting the Research

In this project, the techniques used are general, well known and validated in the literature of the field of NLP and machine learning. The methods like tokenization, stop word removal, text cleaning, "TF-IDF (Term Frequency – Inverse Document Frequency)" vectorization are widely used in the field of text mining and document classification for NLP. These are proven techniques for converting text to data and can be used to analyse using machine learning.

"Naïve Bayes, Support Vector Machine (SVM), Random Forest and XGBoost" algorithms are popular classification algorithms in the field of fraud detection studies and are being used for this study. They have been well tested in a variety of fields such as cybersecurity, spam detection, fraud detection in finance, and identification of online scams, to name a few.

Moreover, the techniques of Explainable AI, such as SHAP and LIME, have gained a lot of interest in scientific research and industrial practice to tackle transparency and accountability concerns in AI systems. These may help researchers and practitioners understand what features are most pertinent to consider when making a classification decision and show an extra advantage for high-stakes usage, such as fraud detection.

5.6 Industry Relevance and Potential Impact

This project will most likely be of enormous interest outside of the academic world. The problem of false employment postings is a problem that involves employment sites, human resource (HR) departments, staffing firms and cyber security companies. A fraud detection tool powered by AI could be useful to automatically detect fraudulent job postings before they are published on the recruitment site. These systems can lower down on fraud incidents, increase the user trust, boost the platform reputation, and reinforce the cybersecurity measures (Taslim et al. 2025).

The explainability part of the proposed framework also contributes to the relevance to the industry. The acceptance of AI solutions among employers and platform administrators is improved by transparency and explainability of the decisions made by the AI. Therefore, the incorporation of Explainable AI has the potential to enhance the trust in automated fraud detection systems and facilitate the responsible use of AI in recruitment settings.

5.7 Summary

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Appendix B: GitHub Link

<https://github.com/mohamedyounus1999-lab/Younus-Project-capstone.git>

Appendix C: Video Link

[Meeting with MOHAMED YOUNUS SHAHUL HAMEED \(Student\)-20260805_133821-Meeting Recording.mp4](#)